SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engine

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	27-07-2026	**Duration:**	60 minutes

Code of Conduct

I V. Tejaswini(192424360) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V. Tejaswini

Annexure A – Agile Sprint Planning & User Story Workshop

Smart Waste Collection Monitoring and Route Optimization Platform

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.

Project Vision

The **Smart Waste Collection Monitoring and Route Optimization Platform** aims to modernize urban waste management by using IoT sensors, GPS technology, cloud computing, and data analytics. The platform continuously monitors the fill level of waste bins, predicts collection needs, and automatically generates optimized routes for waste collection vehicles. This reduces fuel consumption, minimizes operational costs, prevents overflowing bins, and improves city cleanliness.

The platform also enables authorities to monitor waste collection activities in real time, ensuring transparency, accountability, and efficient resource utilization.

Project Objectives

- Monitor garbage bin fill levels using IoT sensors.
- Notify authorities when bins become full.
- Optimize waste collection routes using GPS and shortest-path algorithms.
- Reduce fuel consumption and travel distance.
- Prevent waste overflow and improve public hygiene.
- Provide real-time monitoring dashboards.
- Generate reports and analytics for better decision-making.
- Improve citizen satisfaction through timely waste collection.
- Support smart city initiatives and environmental sustainability.

Project Scope

The project covers:

- Smart bin monitoring
- Route optimization

- Driver navigation
- Administrator dashboard
- Citizen complaint management
- GPS vehicle tracking
- Data analytics and reporting
- Notification and alert system

The project does not include waste recycling plant management or industrial hazardous waste handling.

Key Stakeholders

Stakeholder	Role	Responsibilities
Municipal Corporation	Project Owner	Manage waste collection operations
System Administrator	Platform Management	Monitor system, users, reports
Waste Collection Drivers	Field Operations	Collect waste using optimized routes
Citizens	Service Users	Report overflowing bins and track complaints
Maintenance Team	Technical Support	Repair sensors and devices
Environmental Department	Monitoring	Ensure environmental compliance
IoT Sensor Providers	Hardware Support	Install and maintain smart sensors
Software Development Team	Development	Design, develop and maintain platform

2.Create Epics and User Stories

- Break down the requirements into Epics and formulate well-structured User Stories using the format:

  As a <user>, I want <feature>, so that <benefit>.

- Epic 1: Smart Bin Monitorng

  User Story 1

  As a municipal administrator, I want to monitor the fill level of every waste bin, so that I can schedule waste collection before bins overflow.

  User Story 2

  As a maintenance engineer, I want to receive sensor failure alerts, so that faulty sensors can be repaired quickly.

- Epic 2: Route Optimization

  User Story 3

  As a waste collection driver, I want the shortest optimized collection route, so that I can save fuel and complete collection faster.

  User Story 4

  As a fleet manager, I want vehicle GPS tracking, so that I can monitor driver movement in real time.

  Epic 3: Citizen Services

  User Story 5

  As a citizen, I want to report an overflowing garbage bin, so that the municipality can take immediate action.

  User Story 6

  As a citizen, I want to receive updates on my complaint status, so that I know when the issue is resolved.

  Epic 4: Reports and Analytics

  User Story 7

  As an administrator, I want daily waste collection reports, so that I can evaluate operational performance.

User Story 8

As a city official, I want analytics dashboards, so that I can make better waste management decisions.

2. **Define Acceptance Criteria**

- Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

Acceptance criteria describe the conditions that must be met before a user story is considered complete. They ensure that developers, testers, and stakeholders clearly understand the expected behavior of the system. The Given–When–Then format is used to define these criteria in a structured way.

User Story 1

As a Municipal Administrator, I want to monitor the fill level of every smart waste bin, so that I can schedule waste collection before bins overflow.

Acceptance Criteria

Scenario 1: View Bin Status

- Given the administrator has successfully logged into the system,
- When the administrator opens the dashboard,
- Then the system shall display all waste bins with their current fill levels, location, and status (Empty, Half-Full, Nearly Full, Full).

Scenario 2: Automatic Data Refresh

- Given IoT sensors are sending real-time data,
- When new sensor information is received,
- Then the dashboard shall update the fill level automatically without requiring a page refresh.

Scenario 3: Overflow Alert

- Given a waste bin reaches 90% or more capacity,
- When the sensor sends updated information,

- Then the system shall generate an alert notification for the administrator.

User Story 2

As a Maintenance Engineer, I want to receive alerts when sensors malfunction, so that faulty devices can be repaired immediately.

Acceptance Criteria

- Given a sensor stops sending data for a specified time,
- When the system detects communication failure,
- Then an alert shall be sent to the maintenance team.
- Given the maintenance engineer opens the maintenance dashboard,
- When faulty sensors exist,
- Then all faulty sensors shall be displayed with their location and status.

User Story 3

As a Waste Collection Driver, I want the system to generate the shortest collection route, so that I can reduce travel time and fuel costs.

Acceptance Criteria

- Given multiple bins require collection,
- When the driver begins the trip,
- Then the trip shall be marked as completed.

User Story 4

As a Fleet Manager, I want to track garbage collection vehicles in real time, so that I can monitor operational efficiency.

Acceptance Criteria

- Display live vehicle locations.
- Update GPS location every 30 seconds.
- Show vehicle status (Moving, Idle, Collecting Waste).

- Store travel history for future analysis.

User Story 5

As a Citizen, I want to report overflowing garbage bins, so that authorities can respond quickly.

Acceptance Criteria

- Citizens can submit complaints using mobile or web application.
- Complaint must include location and optional image.
- Complaint ID shall be generated automatically.
- Administrator receives notification immediately.
- Citizen can check complaint status anytime.

User Story 6

As a Citizen, I want to receive complaint status updates, so that I know when the issue has been resolved.

Acceptance Criteria

- Complaint status should change from:
  - Submitted
  - Assigned
  - In Progress
  - Resolved
- Citizens receive SMS or email notification after completion.

User Story 7

As an Administrator, I want daily operational reports, so that I can evaluate waste collection performance.

Acceptance Criteria

- Reports shall display:

- Number of bins collected
- Number of overflow incidents
- Distance travelled
- Fuel consumption
- Collection completion rate
- Average response time

- Reports shall be downloadable in PDF and Excel format.

User Story 8

As a City Official, I want analytical dashboards, so that I can make strategic decisions regarding waste management.

Acceptance Criteria

- Dashboard displays charts and graphs.
- Historical data comparison available.
- Waste generation trends shown monthly.
- Dashboard updates automatically.

Prioritize the Product Backlog

- Organize and prioritize the product backlog based on business value, customer needs, and project dependencies

The Product Backlog is an ordered list of all features, improvements, and technical tasks required for the project. High-priority items are developed first because they provide maximum business value and are essential for the platform's core functionality.

The Product Backlog is a prioritized list of all the features, enhancements, bug fixes, and technical tasks required to develop the Smart Waste Collection Monitoring and Route Optimization Platform. It is maintained by the Product Owner and is updated throughout the project based on stakeholder feedback and changing business requirements. The backlog helps the development team focus on delivering the most valuable features first while ensuring that project dependencies are properly managed.

Backlog Prioritization

The product backlog is prioritized based on business value, customer needs, technical dependencies, and implementation complexity. Features that are essential for the core functionality of the platform are given the highest priority, while additional enhancements are scheduled for later sprints.

High-Priority Items

- User registration and login
- Smart bin monitoring
- IoT sensor integration
- Overflow notification system
- GPS vehicle tracking
- Route optimization
- Administrator dashboard

These features are necessary for the basic functioning of the platform and provide the greatest value to users.

Medium-Priority Items

- Citizen complaint management
- Complaint status tracking
- Daily and weekly reports
- Notification management
- Driver activity monitoring

These features improve the usability and efficiency of the platform but depend on the completion of the core modules.

Low-Priority Items

- AI-based waste prediction
- Advanced analytics dashboard

- Carbon emission reports
- Multi-language support
- Smart recycling recommendations

These features provide additional value and can be implemented in future sprints after the core system is fully operational.

Plan the Sprint Backlog

- Select high-priority user stories for the first sprint and divide them into actionable development tasks.

The Sprint Backlog is a detailed plan of the work that the development team will complete during a sprint. It is created during the Sprint Planning meeting by selecting the highest-priority user stories from the Product Backlog. For the **Smart Waste Collection Monitoring and Route Optimization Platform**, the first sprint focuses on developing the core features required for smart waste management. The sprint duration is **2 weeks**, and the team will work on building a functional prototype that demonstrates real-time monitoring, alerts, GPS tracking, and route optimization.

Sprint 1 User Stories

- User login and authentication
- Smart bin monitoring
- Overflow notification system
- GPS vehicle tracking
- Route optimization
- Basic administrator dashboard

Development Tasks

- Design and develop the login module.
- Integrate IoT sensors with smart waste bins.
- Display real-time bin fill levels on the dashboard.
- Generate alerts when bins reach maximum capacity.

- Integrate GPS for vehicle tracking.
- Develop the shortest route optimization feature.
- Test all modules and fix identified bugs.

Expected Sprint Outcome

By the end of Sprint 1, the team should deliver a working system that allows administrators to monitor smart bins, receive overflow alerts, track collection vehicles, and provide optimized routes for waste collection

Estimate Effort Using Story Points

- Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

Effort estimation is an important step in Agile Scrum that helps the development team understand how much work is required to complete each user story. Instead of estimating work in hours or days, Scrum uses **Story Points** to measure the relative effort, complexity, and uncertainty of a task. This method helps the team plan each sprint effectvely and complete the selected work within the given time.

For the **Smart Waste Collection Monitoring and Route Optimization Platform**, the team uses the **Fibonacci sequence (1, 2, 3, 5, 8, 13)** to assign story points. Tasks with fewer story points are easier and require less effort, while tasks with higher story points are more complex and need additional development, testing, and integration.

Story Point Estimates

- **User Login and Authentication –** This feature involves creating a secure login system with basic user validation. It is a simple task with low complexity.
- **Smart Bin Monitoring –**This is one of the most complex features because it requires integrating IoT sensors, processing real-time data, and displaying the information on the dashboard.
- **Overflow Notification System –** This feature requires detecting when bins are full and automatically sending alerts to administrators. It involves moderate development and testing.
- **GPS Vehicle Tracking –** This module integrates GPS technology to display the live location of waste collection vehicles. It requires map integration and continuous location updates.

- **Route Optimization -** This feature is highly complex because it uses optimization algorithms and GPS data to calculate the shortest and most efficient collection routes.

Why Story Points Are Used

Story points help the Scrum team estimate work more accurately by focusing on task complexity instead of time. They improve sprint planning, balance the team's workload, reduce estimation errors, and make it easier to track progress. Using story points also encourages team discussion and helps deliver project features within the planned sprint duration.

Present the Sprint Goal and Expected Deliverables

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

The Sprint Goal provides a clear objective for the development team during the sprint. It helps ensure that all team members focus on delivering the most valuable features. For the **Smart Waste Collection Monitoring and Route Optimization Platform**, the goal of the first sprint is to develop the essential modules needed for efficient waste collection and monitoring. These features will form the foundation of the system and allow stakeholders to evaluate the initial working version of the platform.

Sprint Goal

Develop the core functionalities of the platform by implementing user authentication, smart bin monitoring, overflow alerts, GPS vehicle tracking, and route optimization to improve the efficiency of waste collection.

Expected Deliverables

- Secure user login and authentication module.
- Smart bin monitoring dashboard with real-time updates.
- IoT sensor integration for monitoring bin fill levels.
- Automatic overflow notification system.
- GPS-based vehicle tracking module.
- Route optimization feature for efficient waste collection.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation,	Story point estimation is well justified, sprint goal is clear, deliverables	Estimation and sprint goal are mostly appropriate with minor	Estimation or sprint goal lacks justification. Presentation is	Estimation is unrealistic or missing. Sprint goal,

Sprint Goal & Presentation	are realistic, and presentation is professional and well organized.	inconsistencies. Presentation is clear.	adequate but incomplete.	deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25